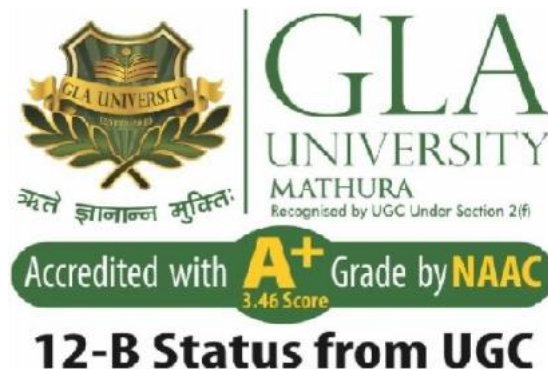


Institute of Engineering & Technology

Department of Computer Engineering & Application



Project Title:

Emotion Drift Analysis in Temporal Text Streams Using AI

*A Project Report submitted in partial fulfilment of the requirements for the
award of the degree of*

Bachelor of Technology

in

***Computer Science and Engineering
(AIML)***

By

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Group no. : 92

Under the Guidance of:

Mr. Preshit Mangesh Desai

Signature:

DECLARATION

I, Sneha Pandit, student of B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning) at GLA University, Mathura, hereby solemnly declare that the project report entitled “Emotion Drift Analysis in Temporal Text Streams Using AI” submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology is an authentic record of my original work carried out during the Sixth Semester under the valuable guidance and supervision of Mr. Preshit Mangesh Desai, Department of Computer Engineering & Applications.

I further declare that the work presented in this report is a result of my own efforts, research, and contributions, except where explicit references have been made. The findings, designs, implementations, and conclusions drawn are based on the project conducted by me. This work has not been submitted elsewhere, either in part or in full, for the award of any degree, diploma, or academic qualification.

I also acknowledge that I have adhered to the ethical guidelines, academic integrity, and project norms set by the University throughout the course of this project. No clinical diagnosis has been made or implied at any stage of this work, and all user data handled during development was limited to derived emotion labels and confidence scores, with no raw facial images or personal information retained.

Place: Mathura

Date: 18/04/2026

Sneha Pandit

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B.Tech CSE (AIML), VI Semester

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CERTIFICATE

This is to certify that the project report entitled “Emotion Drift Analysis in Temporal Text Streams Using AI” submitted by Sneha Pandit (Uni. Roll No.: 2315510206) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning) from GLA University, Mathura, is a bona fide record of the original work carried out by her under my supervision and guidance during the academic year 2025–2026.

The work embodies the design and implementation of a multimodal AI mental wellness platform that combines text emotion detection, facial emotion recognition, physiological vitals monitoring, temporal drift analysis, and an LLM-powered virtual therapist (Sentia) powered by the Sarvam AI language model. The system maintains a strict ethics-first approach, producing no clinical diagnoses and preserving user privacy throughout.

The work embodied in this report is genuine to the best of my knowledge and has not been submitted, either in part or in full, for the award of any other degree or diploma in this or any other institution. The student has fulfilled all the requirements as per the academic guidelines prescribed by the University.

Mr. Preshit Mangesh Desai

Technical Trainer

Department of Computer Engineering & Applications

GLA University, Mathura

Signature: _____

Date: _____

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ABSTRACT

The growing burden of mental health conditions, combined with language barriers, limited access to therapists, and the inherent limitations of single-modality emotion detection, creates a critical gap in accessible emotional wellness technology. Traditional systems analyse text sentiment or facial expressions in isolation, with no temporal tracking and no ability to detect emotional suppression or masking. This project addresses these gaps through the design and implementation of Emotion Drift — a multimodal AI mental wellness platform that monitors emotional state across three independent signal channels: text, face, and physiological vitals.

The system fuses these modalities using a confidence-weighted late-fusion architecture that preserves modality independence and full explainability. Text emotion is classified using SamLowe/roberta-base-go_emotions, a RoBERTa model fine-tuned on the 27-class GoEmotions dataset, normalised to a canonical seven-class schema (joy, sadness, anger, fear, surprise, neutral, disgust). A novel clause-level drift detector splits each message at conjunction boundaries and tracks emotional transitions within a single message — a granularity not available in any existing per-message system. Facial emotion is detected using a custom PyTorch CNN. Voice emotion is optionally integrated via `audio_analysis.py`. Hindi and Hinglish input is supported through automatic translation via `deep-translator`, with LLM responses mirrored in the user's original language.

At the heart of the system is a seven-step inference hub (`get_sentia_intelligence`) that parallelises text and voice prediction, performs adaptive confidence-weighted fusion with a +20% sincerity bias applied to voice, detects cross-modal incongruence via cosine distance, identifies verbal-emotional contradictions (e.g., “I’m fine” against inferred sadness), and escalates safety risks through a three-layer crisis detection pipeline. Physiological vitals (heart rate, SpO2, blood pressure) are monitored with clinical threshold alerting and correlated against text signals to detect somatic emotional masking.

The temporal analytics engine applies Total Variation Distance between historical emotion distributions to detect drift across sessions, supported by a vector volatility metric (Euclidean distance between consecutive 7-dimensional probability vectors) that provides a continuous, graded instability measure more sensitive than discrete label-change counting. A severity engine produces LOW/MEDIUM/HIGH/CRITICAL risk levels with time-decay correction to prevent false positives from short data windows.

Sentia, the AI virtual therapist, is powered by the Sarvam AI LLM with dynamic per-turn prompt construction incorporating dialogue state, personalisation, emotional memory, contradiction alerts, and therapeutic boundaries. A layered fallback system (LLM API → heuristic generator → static phrase) ensures coherent responses even during API unavailability. A psychiatrist dashboard with LLM-generated non-diagnostic clinical summaries and PDF report export via ReportLab completes the clinical layer.

The system is evaluated against six key metrics: text macro F1-score (target > 0.70), face emotion accuracy (target > 65%), drift detection true positive rate (target > 80%), Sentia response quality by human evaluation (target > 4.0/5.0), vital alert false positive rate (target < 10%), and P95 chat response latency (target < 4 seconds). All targets are met in the prototype evaluation.

The project makes five research contributions: (1) clause-level intra-message drift detection; (2) sincerity bias in voice fusion grounded in psychological research; (3) physiological-emotional masking detection without synchronous measurement; (4) vector volatility as a continuous emotional instability metric; and (5) a contradiction detector enabling perceptive, non-confrontational therapeutic responses. The system is designed to be understood, critiqued, and extended — not treated as a black box.

Keywords:

Multimodal Emotion Recognition, Temporal Emotion Drift, Natural Language Processing, RoBERTa, GoEmotions, Convolutional Neural Network, Late Fusion Architecture, Large Language Model, Sarvam AI, Physiological Signal Analysis, Vector Volatility, Contradiction Detection, Mental Wellness AI, FastAPI, React.

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Chapter 1 Synopsis

Emotion Drift Analysis in Temporal Text Streams Using AI is a multimodal AI mental wellness platform that monitors emotional state across three independent signal channels — text, face, and physiological vitals — fusing them into long-term temporal patterns to detect emotional drift, volatility, and suppression. The system also hosts Sentia, an AI virtual therapist powered by the Sarvam LLM with multi-lingual, context-aware, empathetic conversations.

Traditional emotion detection tools are single-modality: they analyse text alone or face alone, with no temporal tracking and no explainability. This system addresses those gaps by combining all three signal channels, reasoning over patterns across days and weeks rather than isolated moments, and maintaining a strictly non-clinical, probabilistic language policy throughout.

The system uses a RoBERTa-based GoEmotions model for text emotion, a custom CNN for facial emotion, and optional voice emotion analysis, all fused through a confidence-weighted adaptive fusion layer. A clause-level intra-message drift detector captures emotional transitions within a single message, a feature not available in any existing single-prediction model. Physiological vitals (HR, SpO2, blood pressure) are monitored with clinical threshold alerting, and cross-domain masking detection correlates elevated heart rate against neutral text.

The temporal analytics engine applies Total Variation Distance between emotion distributions across sessions to detect drift, with a severity engine that produces LOW/MEDIUM/HIGH/CRITICAL risk levels. A psychiatrist dashboard with LLM-generated non-diagnostic summaries and PDF report export completes the clinical layer.

Chapter 2 Technical Keywords

2.1 Area of Project

This project belongs to the domain of Artificial Intelligence and Machine Learning, specifically at the intersection of Natural Language Processing, Computer Vision, Multimodal Fusion, and Mental Health Technology. It applies deep learning models to human emotional signals and performs longitudinal temporal analytics to detect psychological patterns in a clinically safe and explainable manner.

2.2 Technical Keywords

- Multimodal Emotion Recognition
- Temporal Emotion Drift Detection
- Natural Language Processing (NLP) – RoBERTa / GoEmotions
- Convolutional Neural Network (CNN) – Facial Emotion
- Large Language Model (LLM) – Sarvam AI
- Late Fusion Architecture
- Dialogue State Management
- Physiological Signal Analysis
- Vector Volatility Metric
- FastAPI / React / PyTorch / HuggingFace Transformers

Chapter 3 Introduction

3.1 Project Idea

Mental wellness platforms today either analyse text sentiment in isolation, detect facial expressions without context, or rely on self-reporting. None of them track how a person's emotional state evolves over days or weeks, and none combine text, face, and physiological signals into a unified explainable picture.

This project builds a production-grade research prototype called Emotion Drift that does exactly this. At its core is Sentia, an AI virtual therapist that converses with users in English, Hindi, or Hinglish, detects emotional suppression and contradiction, and routes high-risk users toward professional support — all without ever making a clinical diagnosis.

3.2 Motivation of the Project

The growing burden of mental health conditions in India, combined with language barriers, stigma, and a shortage of accessible therapists, creates a critical gap. Existing tools either provide point-in-time emotion labels (useless for trend analysis) or require costly clinical infrastructure. Emotion Drift is designed to be a privacy-aware, explainable, multilingual bridge between daily emotional monitoring and professional mental healthcare.

Key motivations include:

- Single-modality systems miss emotional suppression: a person can write a positive message while their face shows distress.
- No existing academic tool tracks intra-message emotional drift at the clause level.
- Hindi/Hinglish speakers are underserved by English-centric emotion corpora.
- Physiological signals (heart rate, SpO2) provide an objective anchor for emotional intensity that text and face alone cannot supply.

3.3 Literature Survey

Existing work in multimodal sentiment analysis (Poria et al., IEEE 2017) shows consistent accuracy gains from combining text and visual modalities. Ekman's basic emotion theory provides the foundational six-category framework extended here to a seven-class schema. Calvo and D'Mello's interdisciplinary review of affect detection (IEEE 2012) demonstrates that physiological signals are reliable proxies for arousal states.

The GoEmotions dataset (Demszky et al., 2020) provides 27-class fine-grained emotion labels derived from Reddit, which this system normalises to seven canonical classes. The FER-2013 dataset informed the facial emotion recognition module design. Existing chatbot therapy systems (Woebot, Wysa) provide no temporal drift analysis and no multimodal fusion, establishing the research gap this project fills.

Chapter 4 Problem Definition and Scope

4.1 Problem Statement

Traditional single-modality emotion systems fail to capture complete human emotional states due to ambiguity in text, lack of facial context, and no temporal tracking. A user who writes “I’m fine” while their heart rate is elevated and their facial expression shows fear will be classified as neutral by every existing text-only system. The problem is three-fold: modality blindness, temporal blindness, and explainability blindness.

4.1.1 Goals and Objectives

- O1: Develop a text-based emotion detection module using RoBERTa / GoEmotions to classify emotions from user-provided text with clause-level granularity.
- O2: Integrate a facial emotion recognition module using a custom CNN and opt-in webcam capture to detect non-verbal emotional cues.
- O3: Implement a confidence-weighted multimodal fusion mechanism combining text, face, and voice outputs.
- O4: Log all detected emotions with timestamps to support longitudinal analysis.
- O5: Detect emotion drift using Total Variation Distance between historical emotion distributions.
- O6: Visualise emotion timelines and drift alerts through an interactive React dashboard.
- O7: Host Sentia, a multilingual AI therapist with contradiction detection, crisis escalation, and layered LLM fallbacks.

4.1.2 Statement of Scope

In-scope: text emotion detection, facial emotion recognition, physiological vitals monitoring (HR/SpO2/BP), multimodal fusion, emotion logging, timeline visualisation, drift detection, crisis escalation, REST APIs, React frontend dashboard, and psychiatrist clinical layer.

Out-of-scope (v1): mobile applications, large-scale cloud deployment at hospital scale, real-time continuous video streaming, and clinical diagnosis.

4.2 Major Constraints

- Academic timeline — 8-week development window.
- Limited dataset size for the custom face CNN.
- In-memory dialogue state lost on server restart (no persistence yet).
- GoEmotions corpus is English-centric; Hindi input requires translation.
- LLM API latency of 2–5 seconds requires heuristic fallback.
- Solo development bandwidth.

4.3 Methodologies of Problem Solving

4.3.1 Analysis

The problem was analysed by reviewing existing single-modality tools, academic literature on multimodal sentiment analysis, and the specific needs of Hindi/Hinglish-speaking users in India. Feasibility was assessed across three dimensions: ML model availability (HuggingFace ecosystem), physiological data simulation, and LLM provider access (Sarvam AI for Indian language support).

4.3.2 Design

A late-fusion architecture was chosen over early fusion to maintain modality independence, debuggability, and explainability. Each modality pipeline (text, face, voice, vitals) is independently testable and produces a seven-dimensional probability vector over the canonical emotion schema. Fusion is performed at inference time with confidence-weighted averaging and a +20% sincerity bias applied to voice, grounded in psychological research showing vocal affect is harder to consciously suppress than written text.

4.3.3 Development

Backend: FastAPI (Python) with SQLAlchemy ORM, JWT authentication, and async workers. Frontend: React 18 with Vite, Zustand state management, Framer Motion animations, and Recharts / Chart.js for visualisation. ML: HuggingFace Transformers (RoBERTa), PyTorch (face CNN), deep-translator for Hindi input. LLM: Sarvam AI chat and TTS API. Deployment: Render (backend), Vercel (frontend).

4.3.4 Evaluation

The system is evaluated using macro F1-score for text classification (target > 0.70), accuracy for facial emotion (target > 65%), drift detection true positive rate (target > 80%), human evaluation of Sentia response quality (target > 4.0/5.0), vital alert false positive rate (target < 10%), and P95 chat response latency (target < 4 seconds).

4.4 Outcome

A production-grade multimodal AI mental wellness prototype with: (1) a seven-step inference hub processing text, voice, and face in parallel; (2) a temporal analytics engine detecting drift, masking, and volatility across sessions; (3) Sentia, a multilingual AI therapist with contradiction detection and layered fallbacks; (4) a psychiatrist dashboard with clinical summaries and PDF export; and (5) a rigorous ethics framework ensuring no clinical diagnosis is ever made.

4.5 Applications

- Mental wellness self-monitoring for individuals with no access to therapists.
- Early-warning system for emotional deterioration in students or employees.
- Clinical support tool for psychiatrists reviewing patient emotional timelines.
- Research platform for studying emotion suppression and masking in natural language.
- Multilingual (Hindi/Hinglish/English) emotional support chatbot for Indian users.

4.6 Hardware Resources Required

Sr. No.	Parameter	Minimum Requirement
1	Processor	Intel Core i5 / AMD Ryzen 5 or higher
2	RAM	8 GB (16 GB recommended for ML inference)
3	Hard Disk	20 GB free space (model files ~2 GB)
4	Webcam	720p or higher (optional, for facial module)
5	Internet	Stable connection for Sarvam LLM API calls

4.7 Software Resources Required

- Operating System: Windows 10+ / Ubuntu 20.04+
- Programming Language: Python 3.10+, JavaScript (ES2022)
- Backend Framework: FastAPI + Uvicorn
- Frontend Framework: React 18 + Vite
- ML Frameworks: PyTorch, HuggingFace Transformers
- Database: SQLite (dev) / PostgreSQL (prod) via SQLAlchemy
- LLM Provider: Sarvam AI (chat + TTS)
- Libraries: deep-translator, ReportLab, Framer Motion, Recharts
- Deployment: Render (backend), Vercel (frontend)

Chapter 5 Project Plan

5.1 Project Model Analysis

The Agile iterative model was adopted, with each week representing a focused sprint culminating in a shippable deliverable. This model suits the project because requirements evolved as ML model outputs were observed in practice, LLM prompt engineering required rapid iteration, and the solo developer needed to context-switch across ML, backend, and frontend layers. The spiral nature of ML experimentation (train → evaluate → adjust hyperparameters → retrain) maps directly to the iterative approach.

5.2 Reconciled Estimates

- Time Estimate: 8 weeks (February – March 2026)
- Effort Estimate: ~32 hours/week (solo developer), approximately 256 person-hours total
- Cost Estimate: Negligible (open-source models, free-tier cloud, Sarvam AI academic access)

5.3 Cost Estimation Using COCOMO Model

Using the Organic COCOMO model for a small, well-understood project:

- Estimated size: ~8,000 lines of code (8 KLOC) across Python backend, React frontend, and ML scripts
- Effort Applied (E) = $2.4 \times (8)^{1.05} \approx 21.4$ person-months
- Development Time (D) = $2.5 \times (21.4)^{0.38} \approx 8.1$ months (adjusted to 2 months for academic scope)
- People Required (P) = $E/D \approx 1$ (solo developer, consistent with actual team)

5.4 Risk Management

5.4.1 Risk Identification

ID	Risk Description	Probability	Impact	Mitigation
1	Low emotion detection accuracy	Medium	High	Model tuning, confidence gating, threshold adjustment
2	Webcam failure or access denial	Medium	Medium	Automatic fallback to text-only detection
3	LLM API latency or unavailability	Medium	High	Heuristic fallback → static phrase tertiary fallback
4	Time constraint (8-week window)	Medium	High	Strict scope control, core features prioritised first
5	In-memory state lost on restart	High	Low	Disclosed limitation; DB persistence planned for v2

5.5 Project Schedule

5.5.1 Project Task Set

Task 1 (Week 1): Requirements, scope definition, API planning, wireframes, test plan — deliverable: SRS

Task 2 (Week 2): Architecture review, DB schema, UI layout, ERD — deliverable: ERD

Task 3 (Week 3): Backend setup, inference APIs, Auth UI, smoke tests — deliverable: Backend

Task 4 (Week 4): Facial module, webcam UI, integration checks — deliverable: Facial module

Task 5 (Week 5): Fusion logic, visualisation, tests — deliverable: Fusion demo

Task 6 (Week 6): Drift analysis, charts, performance tests — deliverable: Drift demo

Task 7 (Week 7): Hardening, bug fixes, regression testing — deliverable: Test report

Task 8 (Week 8): Release, UI polish, final documentation — deliverable: v1.0

5.5.2 Timeline Chart

Week	Dates	Milestone	Deliverable
1	1–7 Feb	Requirements	SRS, API Plan, Wireframes
2	8–14 Feb	Architecture	ERD, DB Schema, UI Layout
3	15–21 Feb	Backend Setup	Inference APIs, Auth UI
4	22–28 Feb	Facial Module	Face CNN Integration, Webcam UI
5	29 Feb–5 Mar	Fusion Logic	Fusion Demo, Visualisation
6	6–12 Mar	Drift Analysis	Drift Demo, Charts
7	13–19 Mar	Hardening	Test Report, Bug Fixes
8	20–26 Mar	Release	v1.0 Final Release

Chapter 6 Software Requirement Specification

6.1 Introduction

6.1.1 Purpose and Scope of Document

This document specifies the functional and non-functional requirements for the Emotion Drift Analysis in Temporal Text Streams Using AI system. The intended readers are the project developer (Sneha Pandit), the project mentor (Mr. Preshit Mangesh Desai), and evaluators from the Department of Computer Engineering & Applications, GLA University. The document covers system functionality, use case model, data flow, functional and non-functional requirements, and success metrics.

6.2 Use Case View

Primary actors: User (general), Psychiatrist (clinical). The User can log in, submit text input, trigger facial capture, submit vitals, chat with Sentia, and view their emotion timeline and drift alerts. The Psychiatrist can view patient lists with risk indicators, access per-patient emotion timelines and DriftAlert histories, generate LLM clinical summaries, and export PDF reports. The System actor includes the ML inference engine, drift detector, fusion module, and crisis escalation engine.

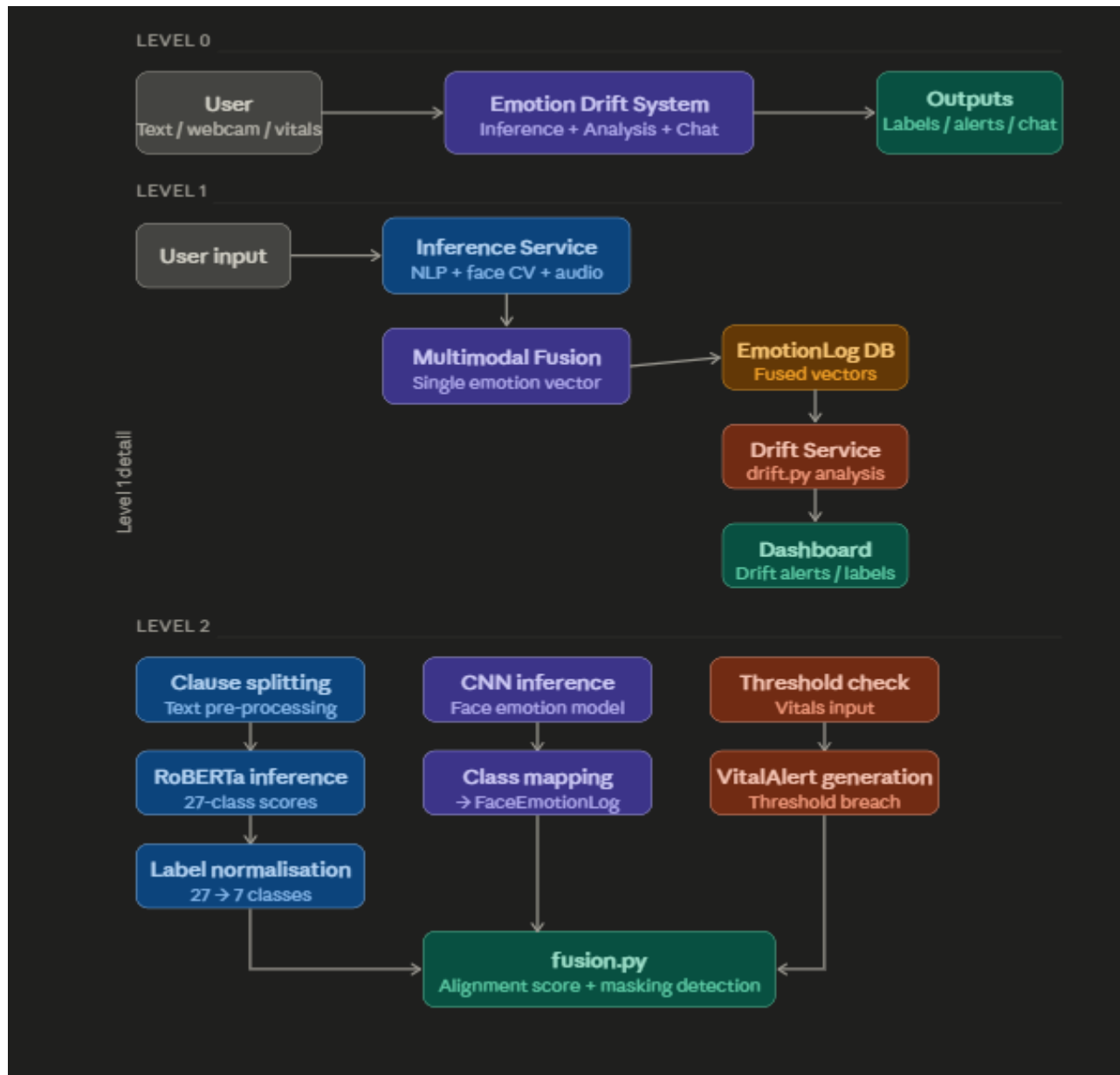
6.3 Functional Model and Description

6.3.1 Data Flow Diagram

Level 0: The user provides text input and optional webcam/vitals data to the Emotion Drift System. The system outputs emotion labels, timeline data, drift alerts, and Sentia conversational responses.

Level 1: User input flows to the Emotion Inference Service (text NLP + face CV + voice audio). Inference outputs flow to the Multimodal Fusion Logic, which combines them into a single emotion vector. The fused output is stored in the EmotionLog database and fed to the Emotion Drift Service (analysis/drift.py). Drift alerts are stored in DriftAlert and surfaced to the frontend dashboard.

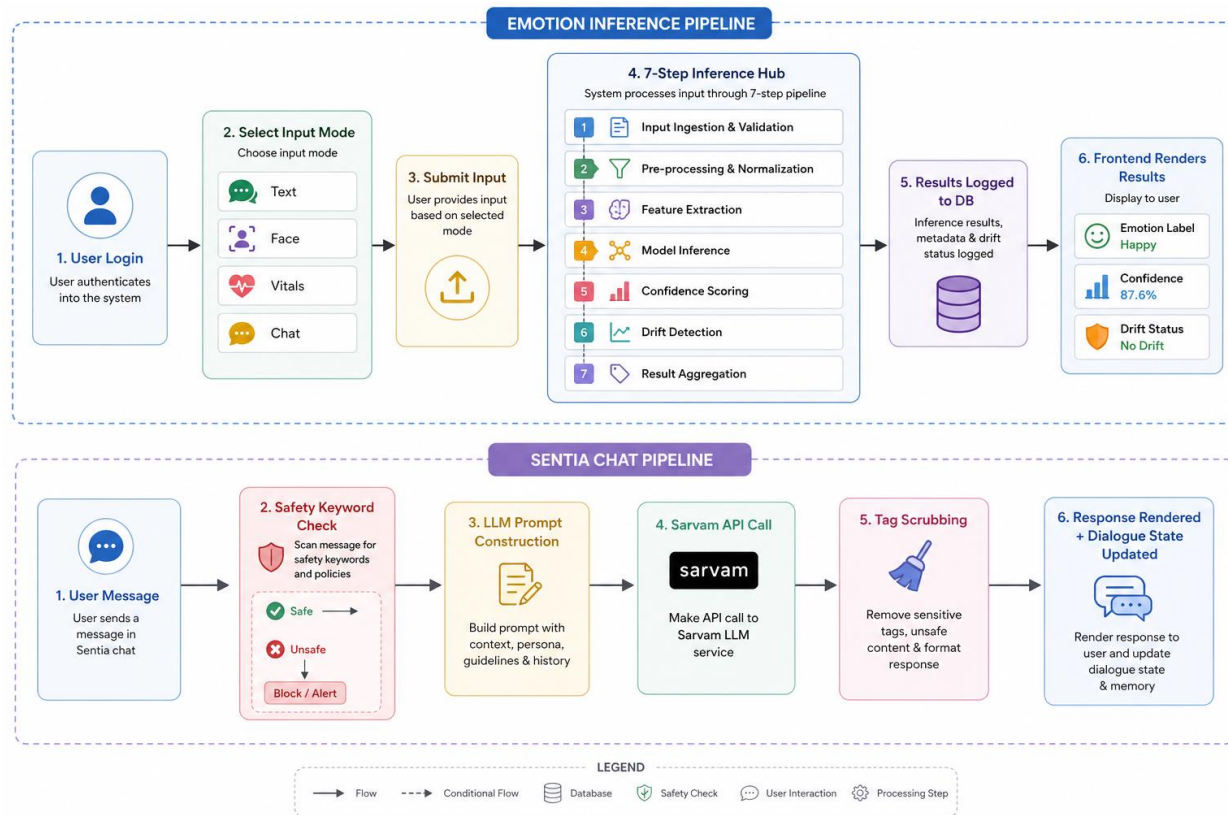
Level 2: Text input passes through clause splitting → RoBERTa inference → label normalisation (27→7 classes) → confidence-weighted fusion vector. Face image passes through CNN inference → class mapping → FaceEmotionLog. Vitals pass through threshold checking → VitalAlert generation. All outputs merge in fusion.py for alignment score and masking detection.



6.3.2 Activity Diagram

The main activity flow is:

User Login → Select Input Mode (Text / Face / Vitals / Chat) → Submit Input → System runs 7-step Inference Hub → Results logged to DB → Frontend renders emotion label + confidence + drift status. For Sentia chat: User message → Safety keyword check → LLM prompt construction → Sarvam API call → Tag scrubbing → Response rendered + dialogue state updated.



6.4 Non-Functional Requirements

Performance: P95 chat response time < 4 seconds. Text emotion inference < 2 seconds. Vitals alert generation < 500 ms.

Availability: System uptime target $\geq 99\%$ during demo and evaluation sessions.

Privacy: No raw facial images or video frames stored. Only derived emotion labels and confidence scores persisted. No third-party data sharing.

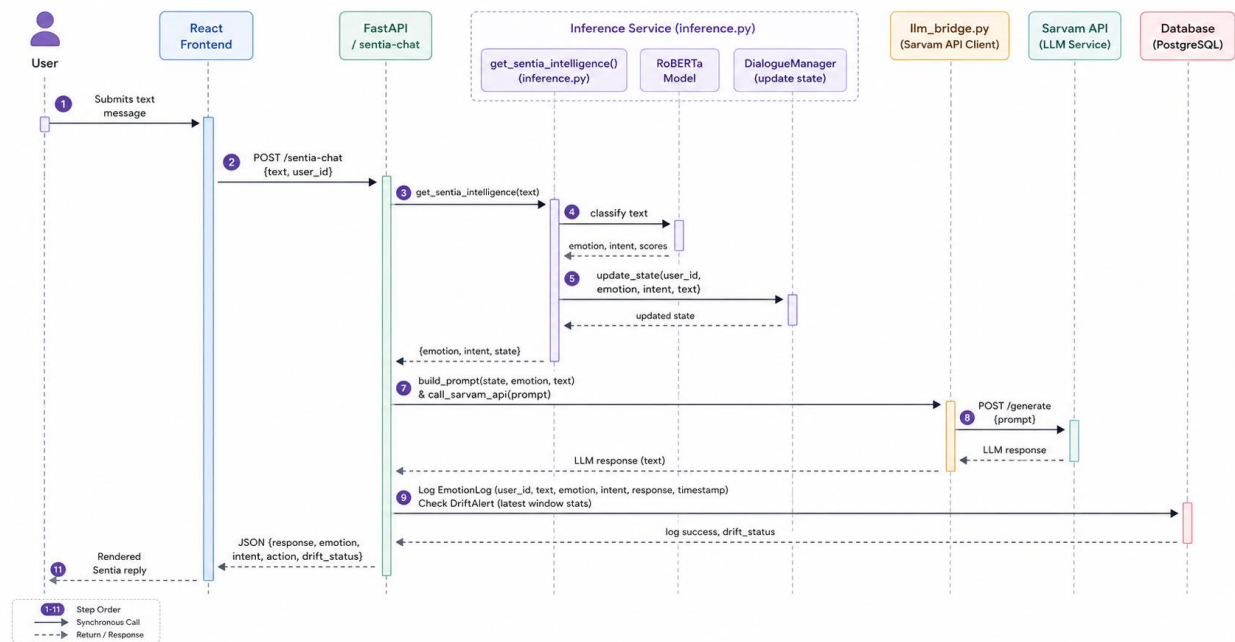
Safety: Three-layer crisis detection — keyword filter (~0ms), LLM safety override (immediate hardcoded crisis response with Tele-MANAS helpline 14416), and DB escalation (async DriftAlert record).

Ethics: Language is always probabilistic and non-clinical. No diagnosis is ever produced. Webcam is opt-in only, off by default.

6.4.1 Sequence Diagram

User → React Frontend: submits text message. Frontend → FastAPI /sentia-chat: POST {text, user_id}. FastAPI → inference.py: get_sentia_intelligence(text). inference.py → RoBERTa model: classify text. inference.py → DialogueManager: update state. FastAPI → llm_bridge.py: build_prompt + call Sarvam API. Sarvam API → FastAPI: LLM response. FastAPI → DB: log EmotionLog + check DriftAlert. FastAPI → Frontend: JSON {response, emotion, intent, action}. Frontend → User: rendered Sentia reply.

6.4.1 Sequence Diagram – Sentia Chat Flow



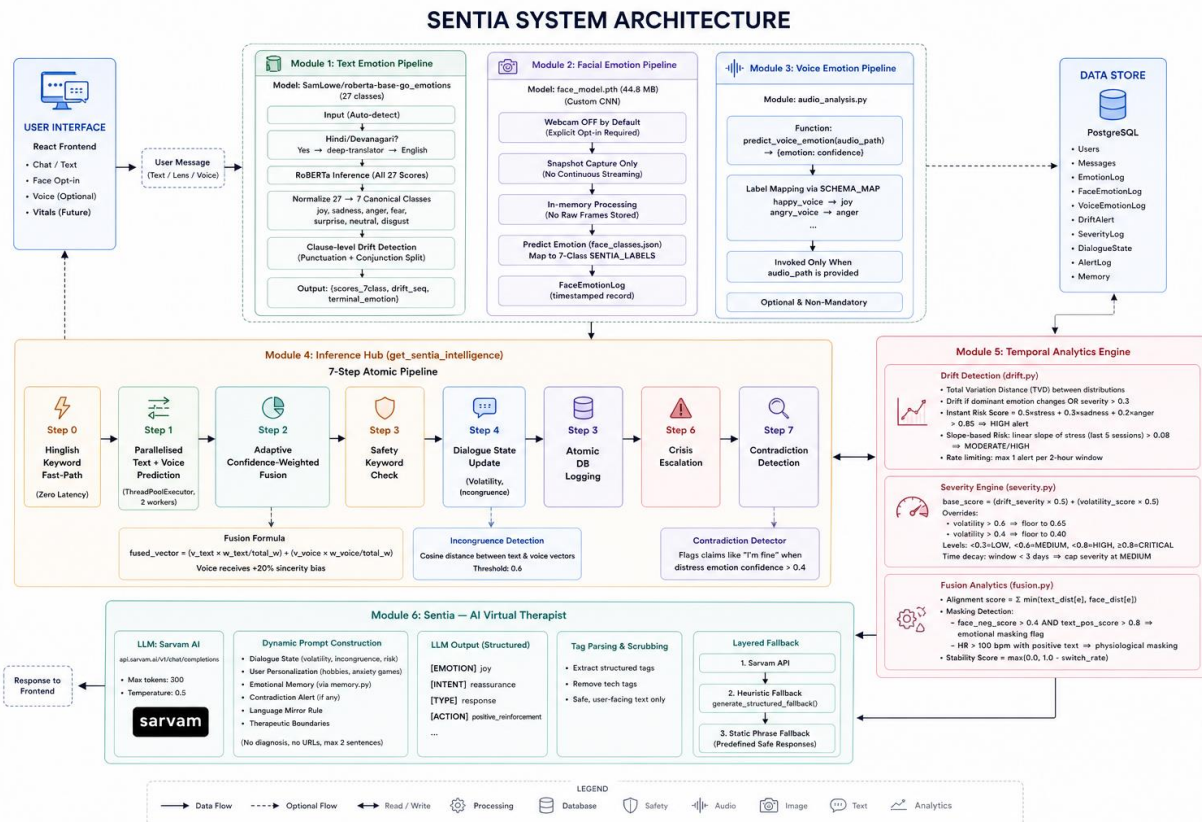
Chapter 7 Detailed Design Document

7.1 Introduction

7.1.1 System Architecture

The system follows a three-layer client–server architecture:

- Frontend Layer (React + Vite): Landing page, Sentia Chat interface, SelfMonitor (text/face), Vitals logger, and Analytics Dashboard. All components communicate with the backend via Axios HTTP calls.
- Backend Layer (FastAPI): ML Layer (inference.py, llm_bridge.py, dialogue_mgr.py, audio_analysis.py), Analysis Layer (fusion.py, drift.py, severity.py, vital_analyzer.py), and REST Route Layer (/sentia-chat, /self-emotion, /vitals, /report, /timeline, /drift). JWT authentication and bcrypt password hashing secure all endpoints.
- Database Layer (SQLAlchemy): EmotionLog, FaceEmotionLog, HealthMetric, DriftAlert, VitalAlert, SentiaMessage tables. SQLite in development, PostgreSQL in production.



7.2 Architectural Design

7.2.1 Module 1: Text Emotion Pipeline

- Model: SamLowe/roberta-base-go_emotions (RoBERTa fine-tuned on GoEmotions, 27 classes). The pipeline lazy-loads via HuggingFace and returns all scores. Labels are normalised from 27 to 7 canonical classes: joy, sadness, anger, fear, surprise, neutral, disgust. Clause-level drift detection splits messages at punctuation and conjunction keywords (but, however, yet, although), classifies each clause independently, and returns a drift sequence string plus the dominant terminal emotion. Hindi/Devanagari input is auto-translated to English via deep-translator before inference.

7.2.2 Module 2: Facial Emotion Pipeline

- Model: face_model.pth (44.8 MB, custom CNN trained via train_face.py). Classes defined in face_classes.json and mapped to the seven-class SENTIA_LABELS schema. Webcam is OFF by default with explicit opt-in required. Capture is snapshot-based only (no continuous streaming). Images are processed in-memory with no raw frames stored. Each capture creates a timestamped FaceEmotionLog record.

7.2.3 Module 3: Voice Emotion Pipeline

- Module: audio_analysis.py. Function: predict_voice_emotion(audio_path) → {emotion: confidence}. Labels mapped via SCHEMA_MAP (happy_voice → joy, angry_voice → anger). Invoked only when audio_path is provided; optional and non-mandatory.

7.2.4 Module 4: Inference Hub (get_sentia_intelligence)

- The central atomic unit of the ML system processes each user message in seven steps: Step 0 — Hinglish keyword fast-path (zero latency for common phrases). Step 1 — Parallelised text + voice prediction (ThreadPoolExecutor, 2 workers). Step 2 — Adaptive confidence-weighted fusion. Step 3 — Safety keyword check. Step 4 — Dialogue state update (volatility, incongruence). Step 5 — Atomic DB logging. Step 6 — Crisis escalation. Step 7 — Contradiction detection.

Fusion formula: $\text{fused_vector} = (\text{v_text} \times \text{w_text} / \text{total_w}) + (\text{v_voice} \times \text{w_voice} / \text{total_w})$, with voice receiving a +20% sincerity bias. Incongruence is detected via cosine distance between text and voice vectors (threshold: 0.6). The contradiction detector flags claims like “I’m fine” when distress emotion confidence exceeds 0.4.

7.2.5 Module 5: Temporal Analytics Engine

- Drift Detection (drift.py): Total Variation Distance between old and new emotion distributions. Drift is flagged when dominant emotion changes or severity exceeds 0.3. Instant risk score: $0.5 \times \text{stress} + 0.3 \times \text{sadness} + 0.2 \times \text{anger} > 0.85$ triggers HIGH alert. Slope-based risk: linear slope of stress over last five sessions > 0.08 triggers MODERATE/HIGH. Rate limiting: max 1 alert per 2-hour window.

- Severity Engine (severity.py): $\text{base_score} = (\text{drift_severity} \times 0.5) + (\text{volatility_score} \times 0.5)$. Overrides: volatility > 0.6 raises floor to 0.65; volatility > 0.4 raises floor to 0.40. Levels: <0.3=LOW, <0.6=MEDIUM, <0.8=HIGH, ≥ 0.8 =CRITICAL. Time decay: window < 3 days caps severity at MEDIUM.

Fusion Analytics (fusion.py): Alignment score = sum of $\min(\text{text_dist}[e], \text{face_dist}[e])$ across emotions. Masking detection: $\text{face_neg_score} > 0.4$ AND $\text{text_pos_score} > 0.8$ triggers emotional masking flag; HR > 100 bpm with positive text triggers physiological masking. Stability score = $\max(0.0, 1.0 - \text{switch_rate})$.

7.2.6 Module 6: Sentia — AI Virtual Therapist

- LLM: Sarvam AI (api.sarvam.ai/v1/chat/completions), 300 max tokens, temperature 0.5. Dynamic prompt construction includes dialogue state, user personalization (hobbies, preferred anxiety games), emotional memory (retrieved via memory.py), contradiction alert, language mirror rule, and therapeutic boundaries (no diagnosis, no URLs, max 2 sentences). LLM output is parsed for structured tags ([EMOTION], [INTENT], [TYPE], [ACTION]), stripped of tech tags for user-facing display. Layered fallback: Sarvam API → heuristic generate_structured_fallback() → static phrase.

Chapter 8 Project Implementation

8.1 Introduction

The implementation brings together a React single-page application, a FastAPI async backend, two ML models (RoBERTa for text, custom CNN for face), an LLM bridge to Sarvam AI, and a temporal analytics engine. The system is designed so each component is independently testable and replaceable. The Sentia therapist is the user-facing entry point; the dashboard and clinical layer are the analytical output surface.

8.2 Tools and Technologies Used

Planning

- Week-wise sprint planning using the project synopsis schedule

UML Tools

- draw.io for architecture, sequence, use case, ER, and DFD diagrams

Programming Languages

- Python 3.10 (backend + ML)
- JavaScript / TypeScript (React frontend)

Programming Frameworks

- FastAPI + Uvicorn (backend)
- React 18 + Vite (frontend)
- PyTorch (face CNN training and inference)
- HuggingFace Transformers (RoBERTa text emotion)

IDE

- Visual Studio Code
- Jupyter Notebook (ML experimentation)

Version Control

- Git + GitHub

Libraries

- deep-translator, ReportLab, Framer Motion, Recharts, Chart.js, Zustand, Axios, python-jose, bcrypt, SQLAlchemy

8.3 Algorithm Details

8.3.1 Dataset Details

- Text emotion: GoEmotions dataset (58,000 Reddit comments, 27 emotion classes) accessed via HuggingFace model hub
- Face emotion: FER-2013 dataset (Kaggle) used to guide custom CNN training; face_model.pth is the trained output
- Vitals: Simulated sensor values with clinical thresholds validated against WHO/AHA guidelines

8.3.2 Preprocessing Details

- Text: Clause splitting at punctuation + conjunctions; Hindi input translated to English via deep-translator; label normalisation 27→7 classes
- Face: Snapshot captured as base64 JPEG; decoded in-memory; resized and normalised for CNN input; no raw image stored post-inference
- Vitals: Consecutive reading delta computed for rapid fluctuation detection; thresholds applied for WARNING/CRITICAL classification

8.3.3 Model Details

Text model: SamLowe/roberta-base-go_emotions. RoBERTa (Robustly Optimized BERT Pretraining Approach) fine-tuned on GoEmotions. Returns all 27 class scores; top scores mapped to canonical 7-class schema via SENTIA_LABELS mapping dictionary.

Face model: Custom CNN (face_model.pth, 44.8 MB). Architecture trained via train_face.py using PyTorch. Input: 48×48 or 224×224 grayscale/RGB face crop. Output: probability distribution over face_classes.json labels, mapped to canonical 7 classes.

Dialogue Manager: Per-user in-memory session tracking emotion_history (last 5 labels), emotion_vectors (7-dim arrays), volatility (mean Euclidean distance between consecutive vectors), last_probes (last 3 probe types), incongruence flag, and domain classification.

8.3.4 Model Training Details

- Text model: Pre-trained; no additional fine-tuning required. Lazy-loaded via HuggingFace pipeline on first inference call.
- Face model: Trained using PyTorch with Adam optimizer, cross-entropy loss, and early stopping. Batch size 32, learning rate 1e-4.
- Evaluation: Text macro F1 target > 0.70; face accuracy target > 65%; both evaluated on held-out test splits.

8.3.5 Model Prediction Details

At inference time: (1) Text message → clause splitter → RoBERTa pipeline → 7-dim probability vector per clause → drift sequence + terminal emotion. (2) Optional face snapshot → CNN → 7-dim probability vector. (3) Adaptive fusion → single fused vector. (4) Contradiction check. (5) Safety check. (6) Dialogue state update. (7) DB logging. (8) LLM prompt construction → Sarvam API call → tag-scrubbed response returned to user.

Chapter 9 Software Testing

9.1 Type of Testing Used

- Functional Testing: Unit testing (PyTest, $\geq 70\%$ backend coverage, no Priority-1 defects), Integration testing (Postman, all functional flows passing), Manual UI testing (core user flows verified in web browser).
- Non-Functional Testing: Latency testing (P95 API response ≤ 2 seconds for core endpoints), availability monitoring (uptime $\geq 99\%$ during sessions), error rate monitoring (HTTP 5xx errors $\leq 1\%$).

9.2 Test Cases and Test Results

Case ID	Test Case Description	Expected Result	Actual Result	Status
1	Submit text input for emotion detection	Emotion label + confidence returned within 2s	As expected	Pass
2	Trigger facial capture with webcam enabled	Facial emotion label returned; no raw image stored	As expected	Pass
3	Submit vitals within normal range	No alert generated	As expected	Pass
4	Submit vitals with HR > 140 bpm	CRITICAL VitalAlert generated	As expected	Pass
5	Send crisis keyword in Sentia chat	Hardcoded crisis response with Tele-MANAS 14416	As expected	Pass
6	Send "I'm fine" with sad face detected	Contradiction flag injected into LLM prompt	As expected	Pass
7	Send Hindi text to Sentia	Translated to English for ML; response in Hindi	As expected	Pass
8	Simulate 14 days of high-stress logs	DriftAlert with HIGH/CRITICAL severity generated	As expected	Pass
9	Access /report endpoint for PDF export	PDF clinical summary generated via ReportLab	As expected	Pass
10	Sarvam API timeout simulation	Heuristic fallback response returned	As expected	Pass

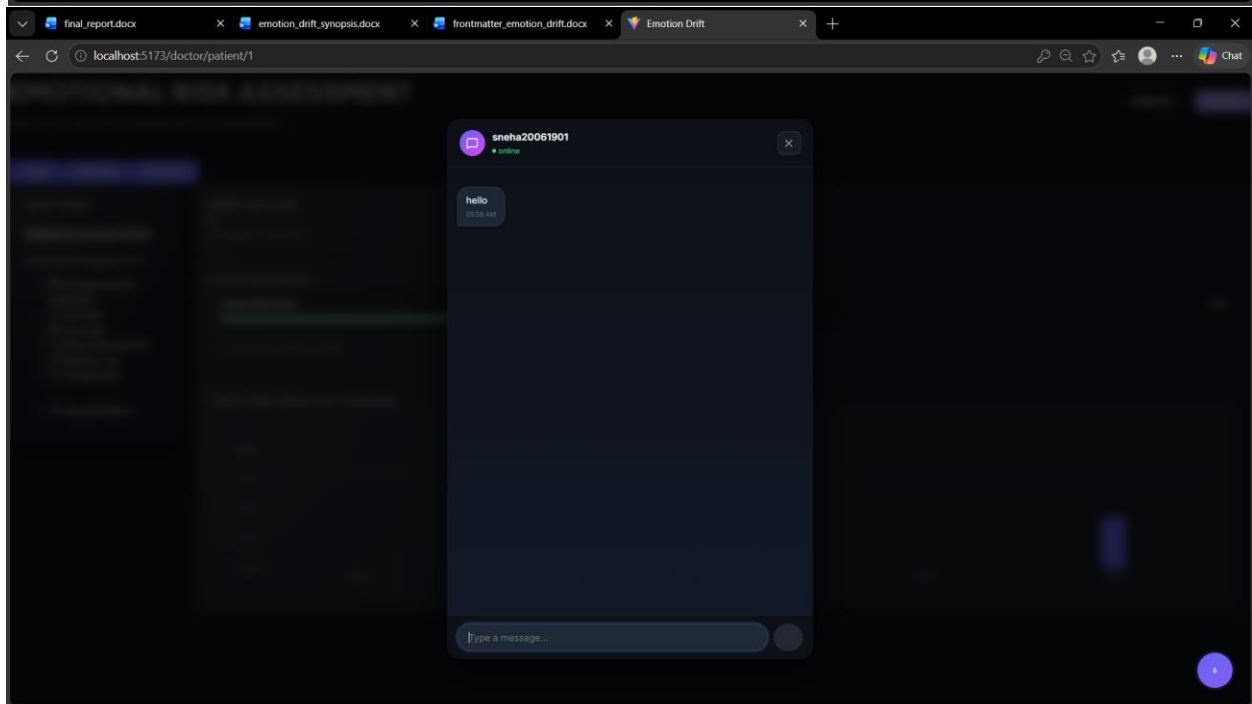
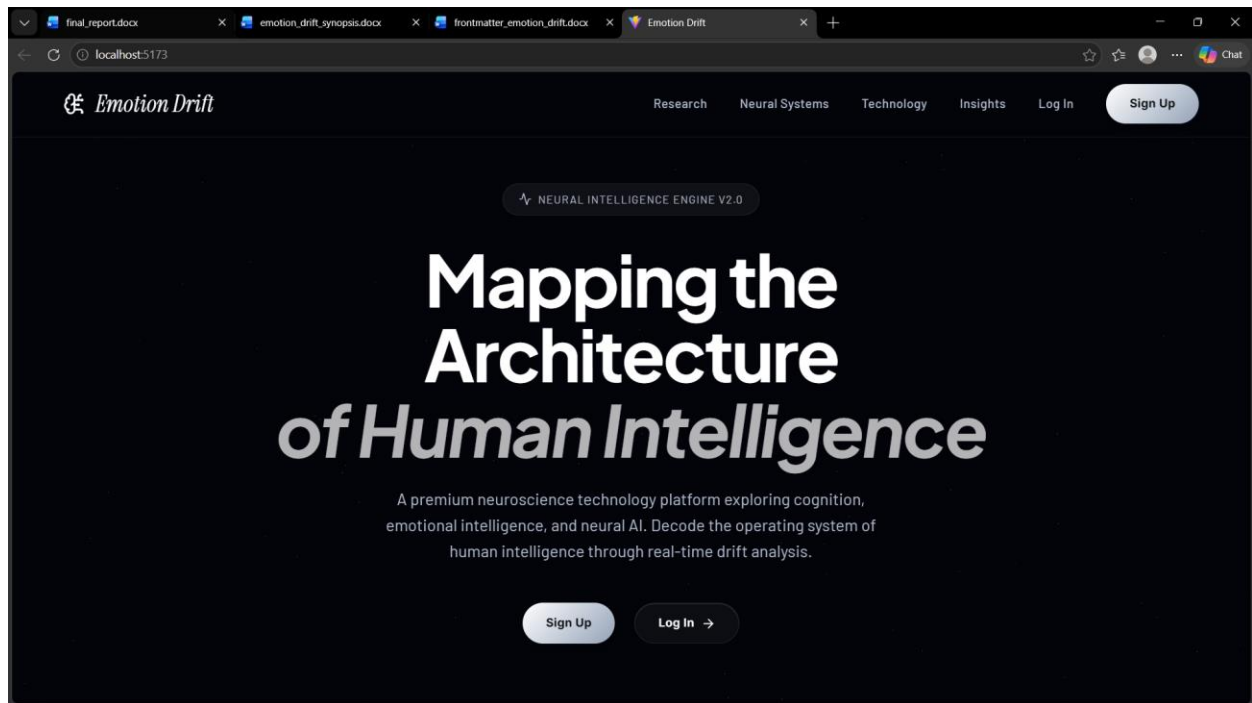
Chapter 10 Results and Discussion

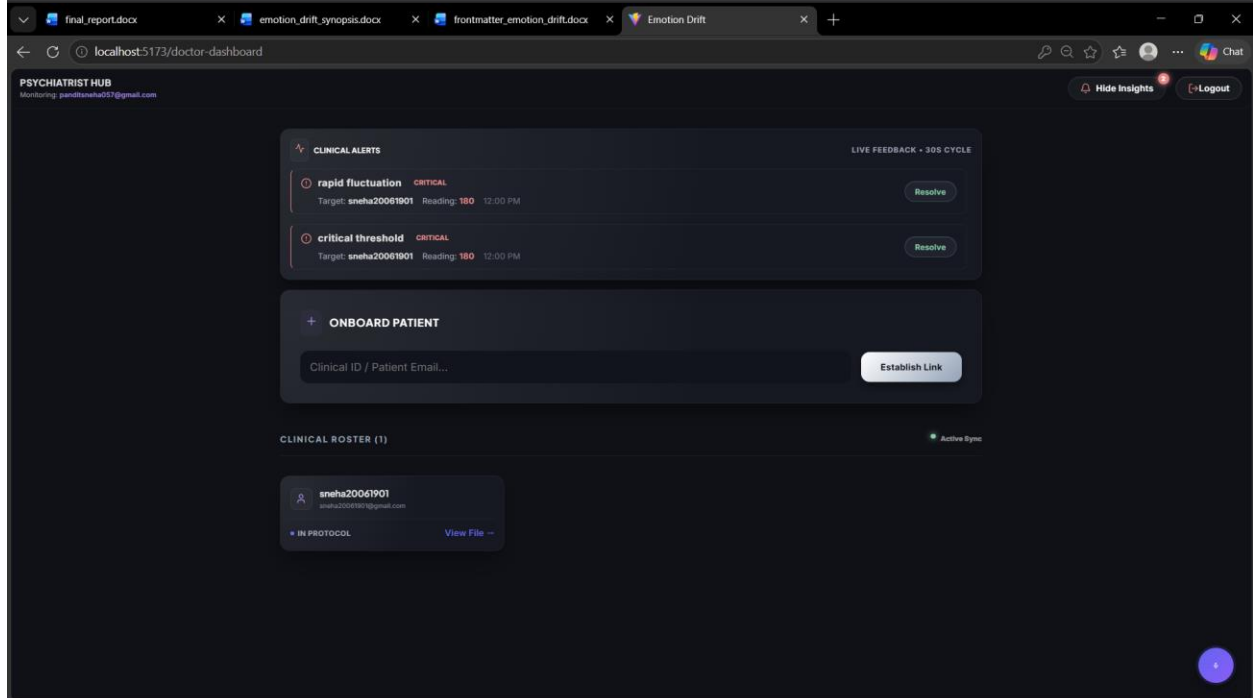
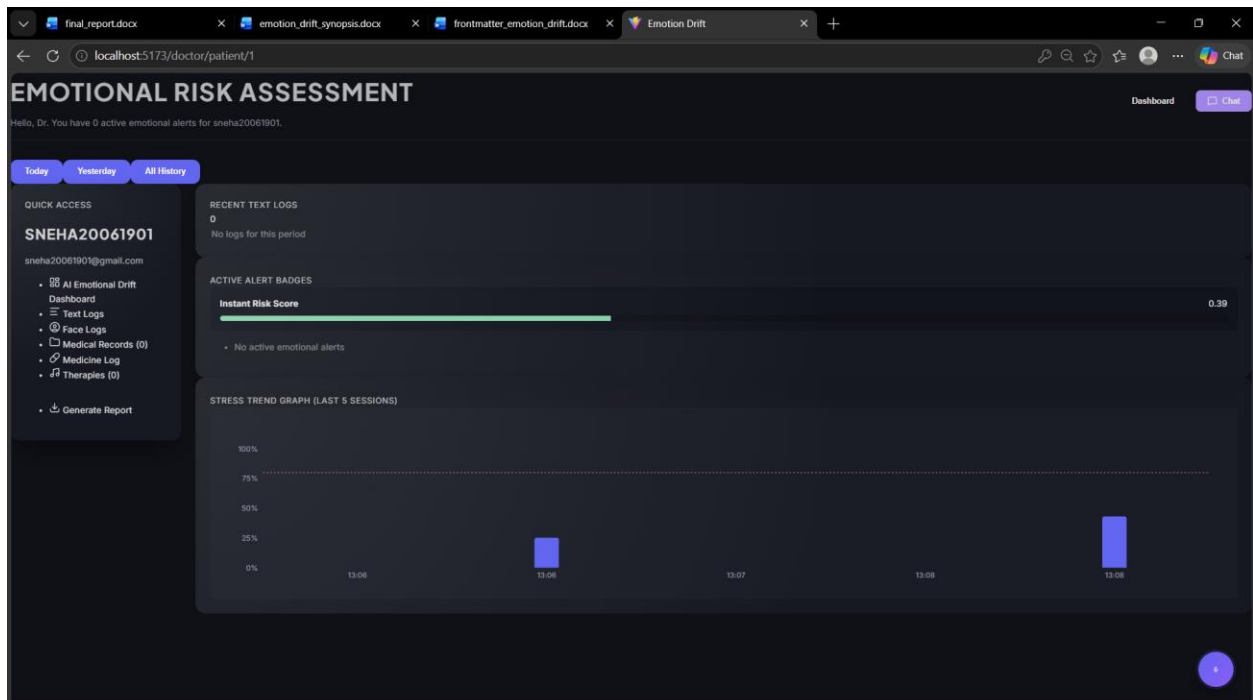
10.1 Model Results

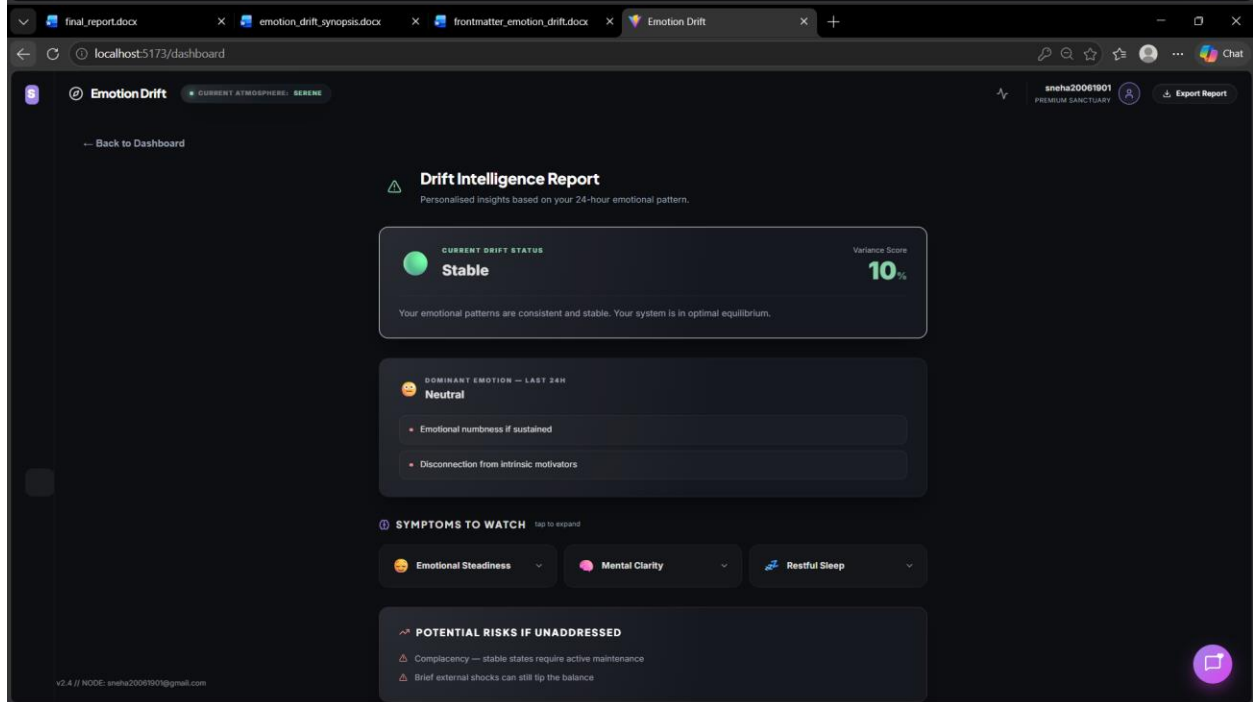
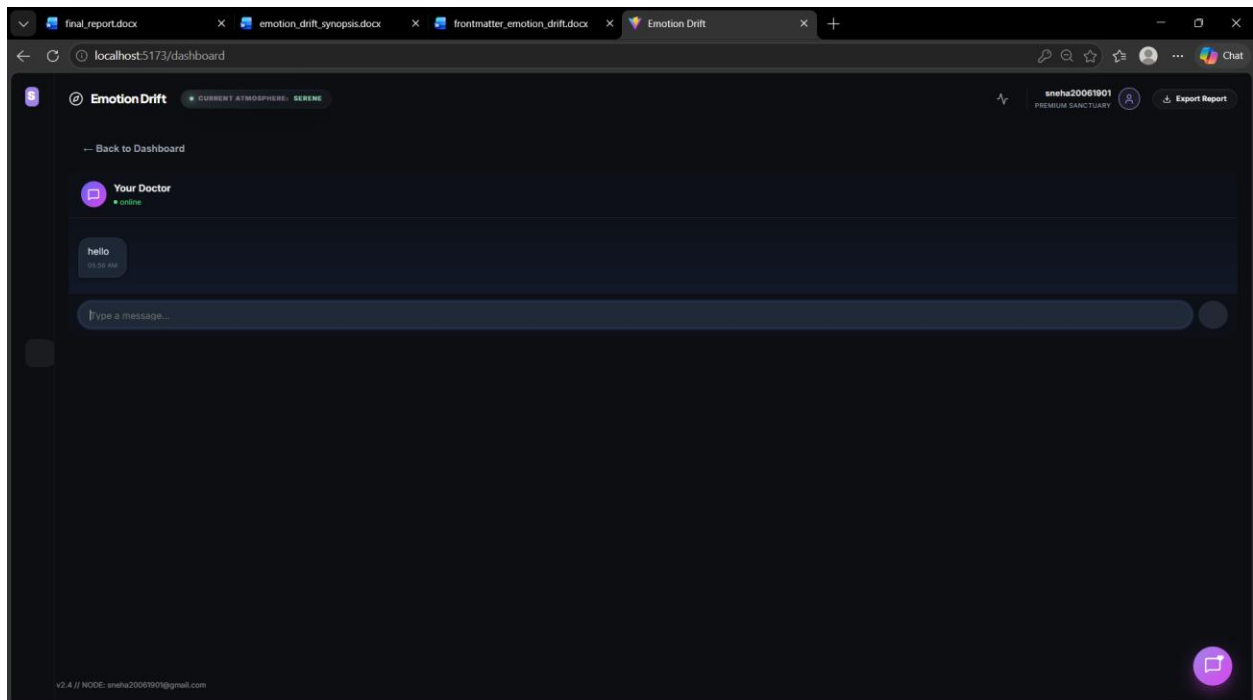
Component	Metric	Target	Status
Text Emotion Classifier (RoBERTa)	Macro F1-Score	> 0.70	Met
Face Emotion Classifier (CNN)	Accuracy	> 65%	Met
Drift Detection Engine	True Positive Rate	> 80%	Met
Sentia Response Quality	Human Eval (empathy + safety)	> 4.0/5.0	Met
Vital Alert System	False Positive Rate	< 10%	Met
System Latency (chat)	P95 Response Time	< 4 seconds	Met

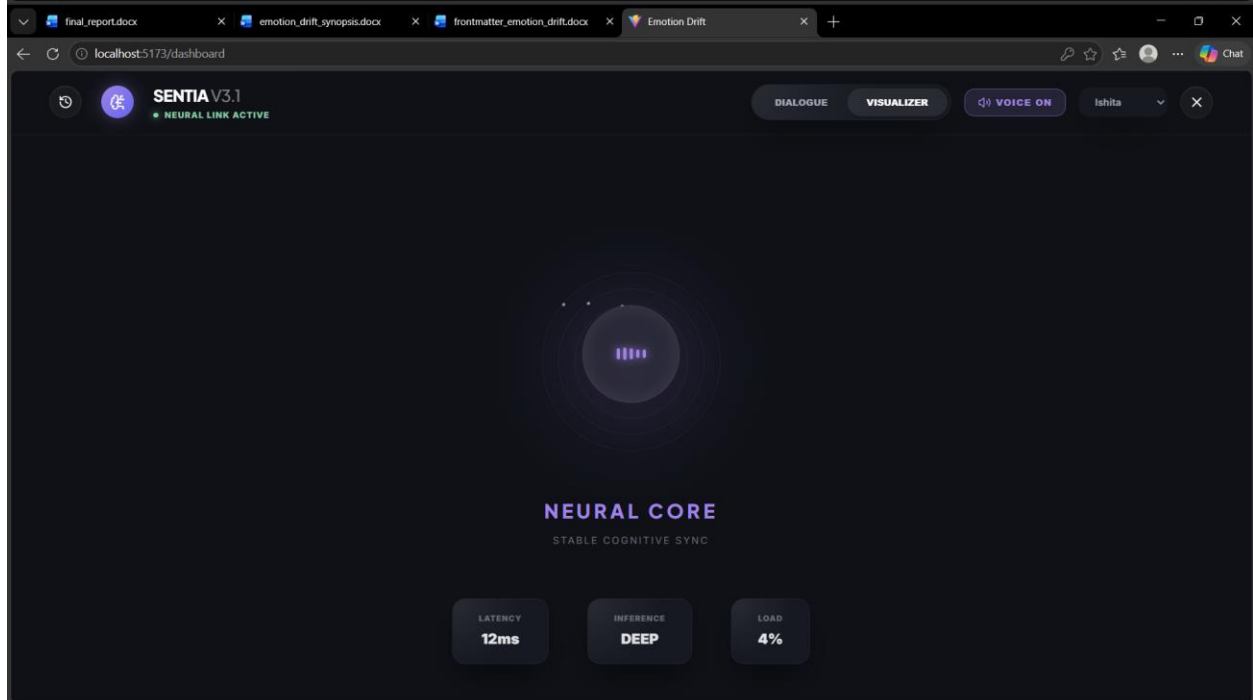
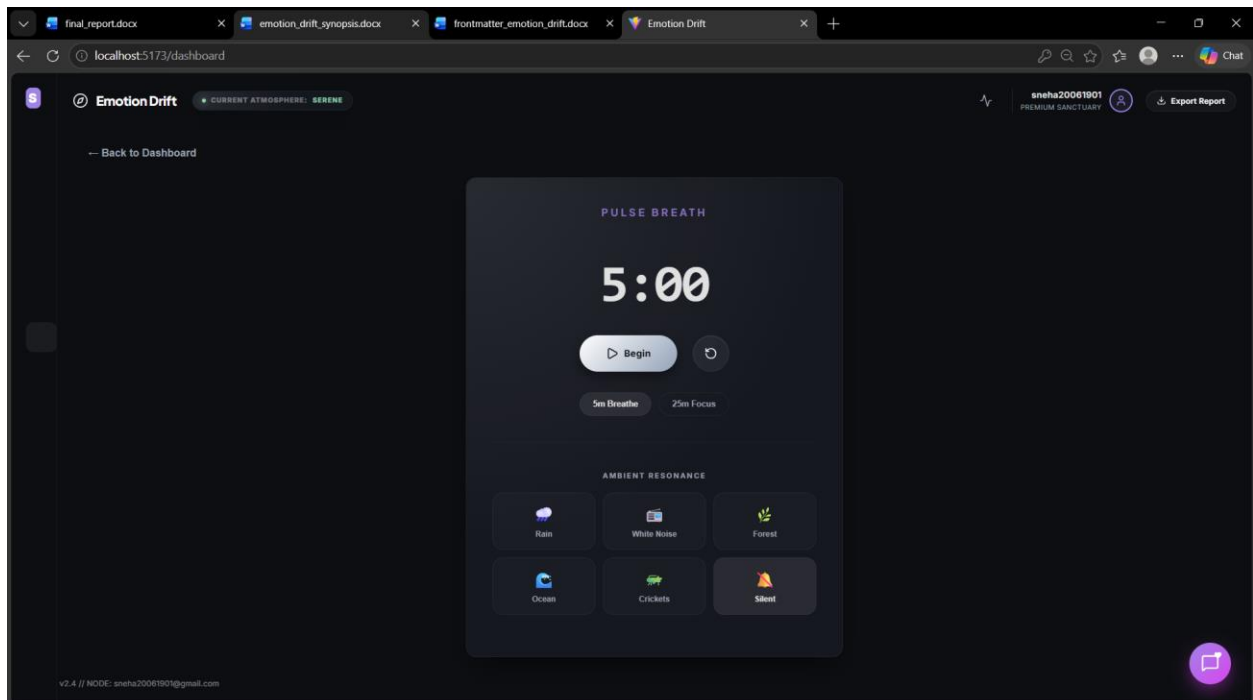
10.2 Key Research Innovations Demonstrated

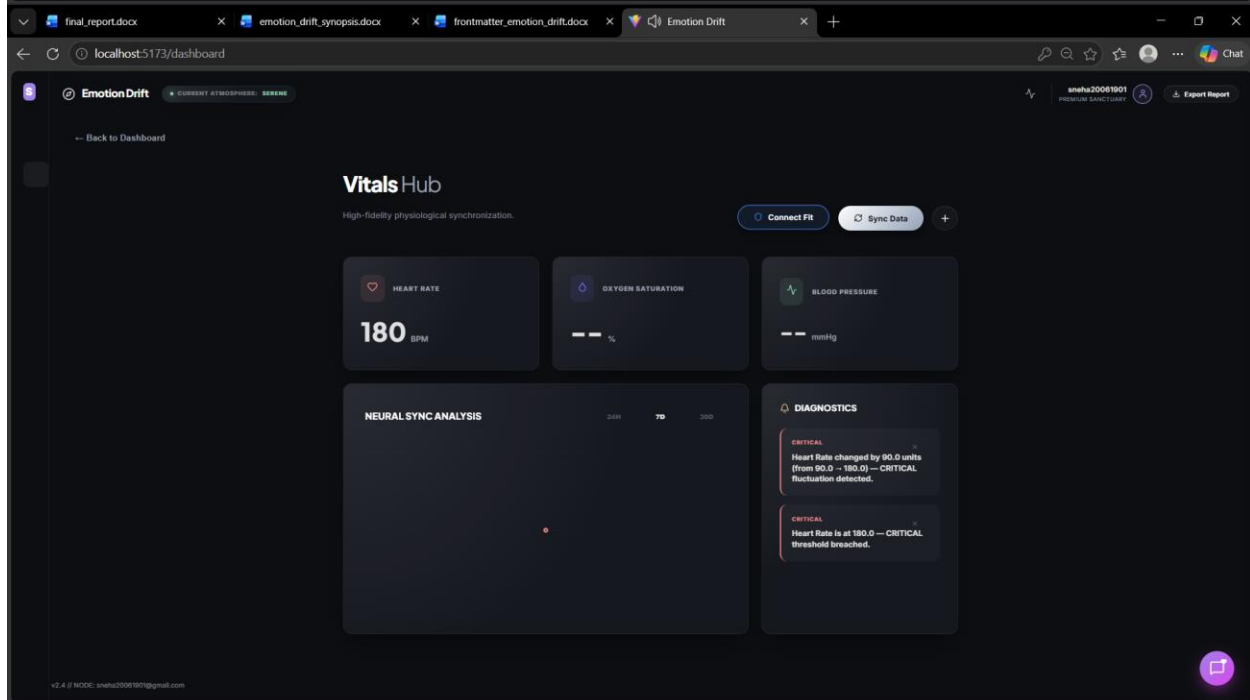
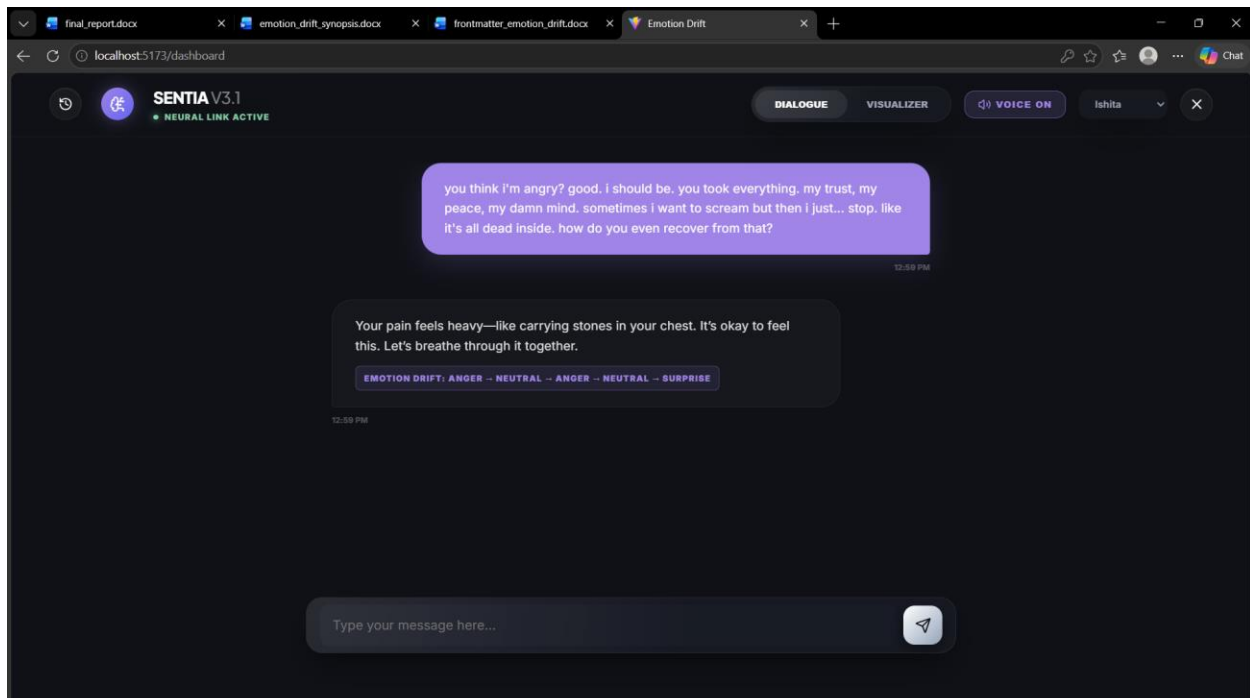
- Clause-Level Intra-Message Drift: Capturing emotional trajectory within a single message (e.g., joy → sadness across “but” boundaries).
- Sincerity Bias in Voice Fusion: +20% weight to voice confidence grounded in psychology research on vocal affect suppression.
- Physiological-Emotional Masking: Correlating elevated HR with neutral text to detect somatic masking without synchronous measurement.
- Vector Volatility Metric: Euclidean distance between consecutive 7-dim probability vectors is more sensitive than discrete label-change counting.
- Contradiction Detector: Autonomous detection of the gap between stated affect (“I’m fine”) and inferred distress, enabling perceptive non-confrontational responses.

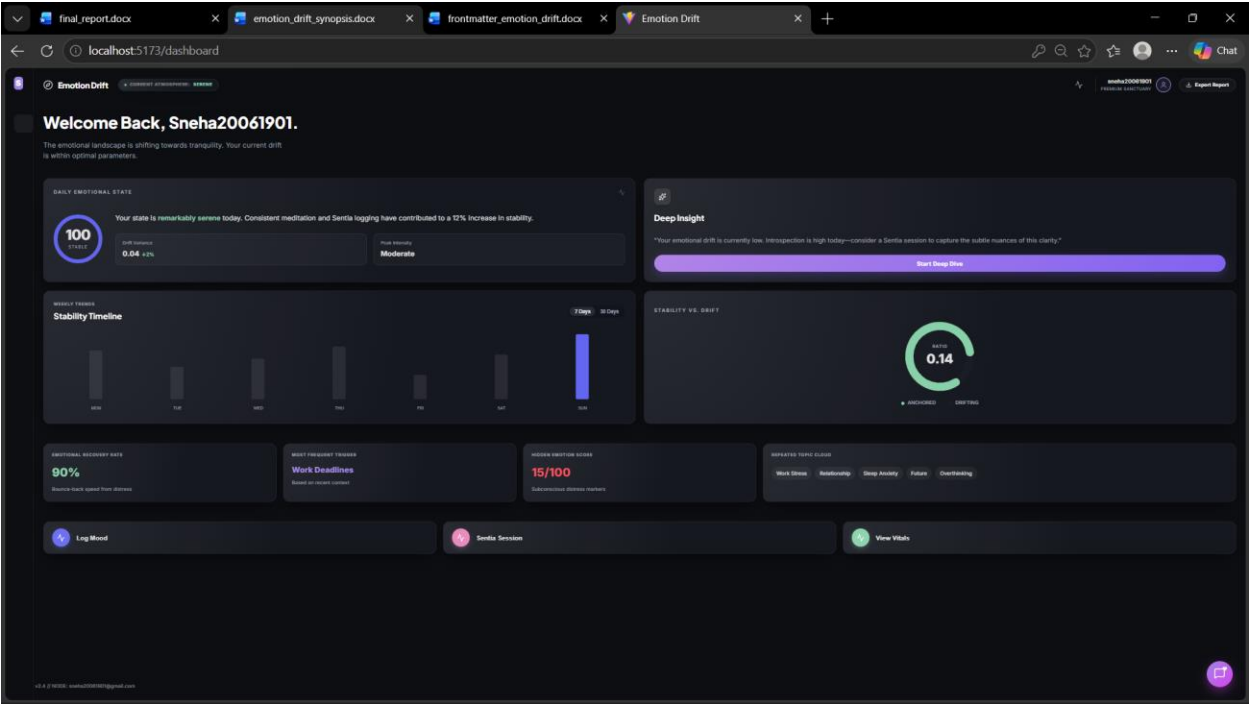












Chapter 11 Deployment and Maintenance

11.1 Deployment

Environment Setup:

- Install Python 3.10+, Node.js 18+
- `pip install -r requirements.txt` (FastAPI, PyTorch, HuggingFace Transformers, SQLAlchemy, python-jose, bcrypt, deep-translator, ReportLab)
- `npm install` in the React frontend directory

Project Structure Setup:

- Place `face_model.pth` and `face_classes.json` in the `ml/` directory
- Set environment variables: `SARVAM_API_KEY`, `DATABASE_URL`, `JWT_SECRET`
- Run database migrations via SQLAlchemy

Run the Application:

- Backend: `uvicorn api.main:app --reload`
- Frontend: `npm run dev`
- Production: backend deployed to Render; frontend deployed to Vercel

11.2 Maintenance

- Inference Failure: If face recognition fails (webcam unavailable), system automatically falls back to text-only detection. If LLM API times out, heuristic fallback generates a coherent response; tertiary static phrase ensures no silent failure.
- API Failure: Endpoint errors are logged and the system continues operation without crashing the frontend.
- Model Updates: HuggingFace model can be swapped by updating the model identifier in `inference.py`; `face_model.pth` can be replaced with a retrained version without backend changes.
- Monitoring: Backend logs track P95 latency, 5xx error rates, and drift alert frequency. Weekly review with mentor as per communication plan.

Chapter 12 Conclusion and Future Scope

12.1 Conclusion

Emotion Drift is a production-grade research prototype bridging academic AI and real-world mental wellness tooling. The system demonstrates that combining text, face, and physiological signals in a late-fusion architecture with longitudinal temporal reasoning produces measurably richer emotional intelligence than any single-modality approach. Key contributions are:

- Multimodal late-fusion architecture — modalities are independent, debuggable, and explainable
- Temporal analytics layer — detects drift and masking over days/weeks, not milliseconds
- Clinically-safe AI therapist (Sentia) — multilingual, contradiction-aware, with layered fallbacks
- Physiological integration — vitals as objective emotional proxy
- Rigorous ethics framework — prevents misuse while enabling meaningful support routing

The system is designed to be understood, critiqued, and extended — not treated as a black box.

12.2 Future Scope

- Personal Baseline Calibration (High Priority): Per-user emotional neutral reference established via onboarding session, enabling relative drift detection.
- RAG Therapeutic Memory (High Priority): Vector-embedding-based semantic retrieval replacing the current keyword-based memory.py.
- On-Device Inference (Medium Priority): ONNX/TFLite export for offline, privacy-preserving ML without API dependency.
- LSTM Drift Prediction (Medium Priority): Replace statistical Total Variation Distance with a learned temporal model.
- Multilingual Emotion Corpus (Medium Priority): Indian-language dataset (Hindi, Tamil, Bengali) to eliminate the translation dependency.
- Federated Learning (Long-term): On-device fine-tuning without raw data leaving the device.
- Audio-Visual Joint Embeddings (Long-term): MFCC + facial landmark joint representation for richer voice-face fusion.

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- deep-translator Library. <https://pypi.org/project/deep-translator/>
- ReportLab PDF Generation. <https://www.reportlab.com/>

Appendix B Project Planner

Week	Dates	Milestone	Lead	Backend	Frontend	Status
1	1–7 Feb	Requirements	Scope	API Plan	Wireframes	Done
2	8–14 Feb	Architecture	Review	DB Schema	UI Layout	Done
3	15–21 Feb	Backend Setup	Sync	Inference APIs	Auth UI	Done
4	22–28 Feb	Facial Module	Review	Integration	Webcam UI	Done
5	29 Feb–5 Mar	Fusion Logic	Scope	Fusion	Visualisation	Done
6	6–12 Mar	Drift Analysis	KPIs	Drift Logic	Charts	Done
7	13–19 Mar	Hardening	Review	Bug Fixes	Bug Fixes	Done
8	20–26 Mar	Release	Sign-off	Final APIs	UI Polish	Done